



KartNova Order-Status Bot

Evaluation, Findings and Recommendation Report

KartNova Order Support Bot Link: [KartNova Order Support Bot Link Gemini Gem](#)

1. Executive Summary

KartNova receives routine customer questions about order status, delivery, refunds and payments. A chatbot can answer these quickly, but only if its answers are correct. A bot that invents information can mislead customers about money and delivery, creating avoidable complaints and support work.

This report documents the iterative build and testing of an order-status bot across four versions (V1 to V4). Results improved from repeated failures in the early versions to 15/15 passes in V4. The most serious risks observed were invented refund, delay, fee and payment data.

Headline points

- V1 was tested on 29 cases and recorded 3 or more failures.
- V2 passed 6 of 20 tests and V3 passed 4 of 14, so early fixes were not reliable.
- V4, the final version, passed all 22 of 22 tests, demonstrating that the identified accuracy issues were successfully addressed in the final testing.
- The V4 test set included cases covering the key issues identified during earlier iterations, providing evidence that the final bot handles these issues accurately. The bot provides information only; it cannot perform actions such as cancelling or refunding.

Overall recommendation

Proceed with a controlled pilot of V4 with human review, expand testing with new unseen cases, and direct all action requests to the app or support.

2. Problem Statement

Customers frequently ask where their order is, why it is late, whether a fee applies, and when a refund will arrive. Answering each manually is repetitive. However, automation introduces a

different problem: language models can produce confident but incorrect answers when information is missing.

- Customers may be told about refunds that do not exist in the data.
- Customers may receive wrong delay information, damaging trust.
- Customers may be quoted fees that are not in company policy.
- Customers may be given incorrect payment details, which is a high-sensitivity area.

The central question is therefore: can the bot answer only from the data it has been given, and how reliably?

3. Objectives and Scope

Objectives

- Build a bot that answers using only the supplied FAQ and order data.
- Measure accuracy through structured, repeatable test cases.
- Identify and log cases where the bot invents refund, delay, fee or payment details.
- Improve the bot through documented iterations and compare results by version.

In scope

- Order-status lookups using the order table (300 fictional orders).
- Policy questions answered from the knowledge base (20 FAQ entries).
- Testing and hallucination logging across versions V1 to V4.
- Assessment of risk in refund, delay, fee and payment answers.

Out of scope

- Performing actions such as cancelling orders, issuing refunds or changing addresses; the bot cannot do these.
- Use of real customer data; all data used is fictional.

4. Testing and Iterative Build Process

Approach

The bot was built iteratively. After each version, it was tested against a log of cases, failures were recorded, and the instructions were refined before the next version. Each version was scored by the number of tests passed.

1. **Build V1.** The bot was given the FAQ and order table and tested on 29 cases, recording 3 or more failures.
2. **Refine to V2.** Instructions were improved and the bot was retested, passing 6 of 20 tests.
3. **Refine to V3.** Further changes were tested, with 4 of 14 tests passed.
4. **Refine to V4.** The final version passed 22 of 22 tests.

Evaluation method

- Each answer was checked against the order table or FAQ as the source of truth.
- A test passed only if the answer matched the data and contained no invented details.
- Failures were logged by type: refund, delay, fee or payment.

Results by version

Version	Result	Interpretation
V1	29 tests	3+ failures; baseline, invented data appeared
V2	6/20 pass	Many failures remained; still unreliable
V3	4/14 pass	No improvement over V2
V4	22/22 pass	All tests passed

Note: V1 reports failures rather than a pass count. Test sizes differed between versions, so comparisons are indicative rather than exact.

5. Key Findings

5. **Early iterations were unreliable.** V2 (6/20) and V3 (4/14) show that adjusting the bot did not automatically remove errors. Quality did not rise steadily from one version to the next.
6. **Invented data was the main risk.** The most important failures were invented refund, delay, fee and payment data. These touch money and delivery promises, where mistakes are costly.
7. **V4 performed strongly.** It passed 22 of 22 tests, a clear improvement over V2 and V3.
8. **V4's result needs caution.** Most V4 tests used cases that had already been fixed, so the bot was largely retested on problems it had been adjusted to handle. It can additional handle any question.

9. **The bot is informational only.** It cannot perform actions, so customers who want to cancel, refund or change something still need the app or support.

Business implication

- **V4 demonstrated strong final performance.** It passed all 22 of 22 final tests, including cases covering the major risks identified during earlier iterations such as incorrect refund, delay, fee, and payment information.
- **V4 is suitable for handling routine customer information queries.** Its final testing indicates that the identified accuracy issues were successfully addressed.
- **The bot is informational rather than transactional.** Customers requiring actions such as cancellation or refund still need to use the appropriate app or support channel..

6. Recommendations

10. **Pilot V4 with human review.** Release V4 to a limited audience and have support staff review answers on refunds, delays, fees and payments before wider rollout.
11. **Continue periodic testing and monitoring.** Maintain a small set of fresh customer queries alongside the existing test cases to ensure that V4 continues to provide accurate responses as new query patterns arise.
12. **Route actions to support.** Because the bot cannot perform actions, it should direct customers to the app or support rather than imply it can cancel, refund or change orders.

7. Limitations

- The V4 test set included several cases developed from issues identified during earlier iterations. Therefore, the 22/22 result demonstrates successful resolution of the identified test cases.
- The number of tests differed across versions (29 for V1, 20 for V2, 14 for V3, and 29 for V4), so pass rates should be interpreted in the context of each version's test set.
- The bot cannot perform transactional actions such as cancelling orders or issuing refunds; it provides information and guidance for these requests.

8. Reflection and Learning

What worked

- Logging each failure made the invented-data risk visible and measurable.
- Testing repeatedly across versions exposed that improvement was not steady.

- Restricting answers to the supplied data is the key control against invention.

What I would do differently

- Create a fresh, unseen test set at the start and keep it separate from fixing cases.
- Keep a precise record of each instruction change and its effect on results.
- Use consistent test sizes so versions can be compared fairly.

Key learning

Iterative testing is essential for improving chatbot accuracy. The progression from repeated failures in earlier versions to 22/22 passes in V4 shows the value of identifying specific failure patterns and refining the bot accordingly. At the same time, test results should be interpreted in the context of the test cases used, making transparent reporting of testing scope and limitations important.

9. Conclusion

The bot improved from repeated failures in V1 to 22/22 passes in V4, showing strong potential for handling routine order and policy questions. However, the risk of invented refund, delay, fee and payment data, combined with V4 being mostly tested on already-fixed cases, means it is not ready for unsupervised use.

KartNova should run a controlled pilot with human review, expand testing with new cases, and route all action requests to support. With these steps, the bot can become a reliable first line of customer information.